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Bihar Engineering University, Patna

End Semester Examination-2023

Time: 03 Hours

Code: 101603 ***Subject: Design of Steel Structure*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[14]**

(a) The dead load includes
- (i) Self wt of the structure
- (ii) All superimposed loads
- (iii) wt of stationary equipment
- (iv) All of the above

(b) The effective throat thickness in case of incomplete penetration butt held is taken as
- (i) $7/8^{th}$ of the thickness of the thicker part joined
- (ii) $7/8^{th}$ of the thickness of the thinner part joined
- (iii) $5/7^{th}$ of the thickness of the thicker part joined
- (iv) $5/7^{th}$ of the thickness of the thinner part joined

(c) The effective length of fillet weld should be less than
- (i) the size of weld
- (ii) two times the size of weld
- (iii) three times the size of weld
- (iv) four times the size of weld

(d) The practical safety factor for the material of bolts is
- (i) 1.00
- (ii) 1.10
- (iii) 1.15
- (iv) 1.25

(e) Which of the following is not a compression member?
- (i) tie
- (ii) strut
- (iii) rafter
- (iv) boom

(f) What is the maximum effective slenderness ratio for a tension member in which reversal occurs
- (i) 180
- (ii) 200
- (iii) 280
- (iv) 300

(g) Lacing shall be designed to resist a total transverse shear equal to of axial force
- (i) 5%
- (ii) 1%
- (iii) 4.3%
- (iv) 2.5%

(h) An ideal section for compression
- (i) one having different moment of inertia about any axis through its centre of gravity
- (ii) one having same moment of inertia about any axis through its centre of gravity
- (iii) one having larger length
- (iv) one made up of costly material

(i) Proof stess for minimum bolt tension is
- (i) $0.7f_{yb}$
- (ii) $0.5f_{yb}$
- (iii) $0.7f_{ub}$
- (iv) $0.5f_{ub}$

(j) The flange is classified as plastic if outstand element of compression flange of rolled section is than
- (i) 8.4ε
- (ii) 9.4ε
- (iii) 10.5ε
- (iv) 15.7ε

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Q.2 Solve this question : **[14]**

Design a lap joint between the two plates each of width 120mm, if the thickness of one plate is 16 mm and the other is 12 mm. The joint has to transfer a design load of 160 kN. The plates are of Fe410 grade. Use bearing type bolt.

Q.3 Solve both questions : **[14]**

(a) List the 8 loading combination to be considered when considering only DL., LL. and EL.

(b) What are the factors that influence wind pressure coefficient?

Q.4 Solve both questions : **[14]**

(a) Write short note on: (i) Plastic Section (ii) Compact Section (iii) Semi-Compact Section

(b) Design a slab base for a column ISHB 350 @ 710.2N/m subjected to the factored axial compressive load of 120kN. Load is transferred to the base plate by direct bearing of Column flanges take f_u= 410 MPa, f_y= 250 MPa.

Q.5 Solve both questions : **[14]**

(a) Calculate the moment carrying capacity of a laterally Unrestrained ISMB 400 member of length 3 m.

(b) Differentiate between local and lateral buckling of beam.

Q.6 Solve this question : **[14]**

Design a built-up laced column with four angles to support an axial load of 900kN. The column is 12 m long and both the ends are held in position and restrained against rotation. Assume Fe 410 grade steel.

Q.7 Solve both questions : **[14]**

(a) A tie bar 100 mm x 16 mm is welded to another plate. It is subject to factored pull of 300 kN. Find the minimum overlap required if 8 mm site fillet welds are used. Assume any missing data.

(b) Why is lacing provided in the columns?

Q.8 Solve this question : **[14]**

A tie member of truss consists of double angle section each 80 mm x 80 mm x 8 mm welded on the opposite side of a 12 mm thick gusset plate. Design a fillet weld for making connection in the workshop. The factored tensile force in the member is 300 kN.

Q.9 Write short notes on any four of the following:- **[14]**

(a) Prying force
(b) Failure modes of bolted joint
(c) Lateral tensional buckling
(d) Intermediate stiffer
(e) Parameter affecting behaviour of beam column
(f) Rolled steel section with suitable diagram

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